

Proveria

Private Provenance for High-Stakes Digital Materials

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Executive Summary

Organizations increasingly need to prove facts about digital materials they cannot safely disclose.

A law firm may need to prove that a contract clause existed in the signed version of an agreement. A financial institution may need to show that a policy was in effect when a decision was made. A pharmaceutical company may need to demonstrate that a research protocol existed at a particular milestone. An AI developer may need to answer whether specific content was present in a training corpus. A government contractor may need to verify the integrity of submitted records. A newsroom may need to defend the authenticity of source materials without exposing its full archive.

Today, these organizations usually face two bad options: disclose sensitive materials to prove the claim, or ask the counterparty to trust internal records.

Disclosure can expose privileged documents, regulated data, trade secrets, confidential research, licensed content, or source material. Internal records can be useful, but they are often controlled by the same organization making the claim. In adversarial settings, that creates a trust gap.

Proveria creates a third option: prove the content-level fact without exposing the content itself.

Proveria lets a producer create a private, cryptographic attestation over selected files and document content from the producer's own machine. The producer keeps the underlying materials. Proveria receives cryptographic metadata, not plaintext content. Later, an authorized verifier can check whether a specific file or passage matches that attested set and receive a durable result package or receipt.

The core idea is simple:

The content stays. The proof travels.

Proveria is built for legal teams, regulated enterprises, AI labs, research organizations, media organizations, government contractors, and any group that needs to answer high-stakes provenance questions without turning verification into disclosure.

1. The Provenance Gap

Digital organizations generate enormous volumes of important material: contracts, policies, source code, research data, training corpora, filings, communications, evidence packages, clinical documents, model datasets, publications, and media archives.

Much of that material becomes important only later, when someone asks a hard question:

- Did this document exist in this form on this date?
- Was this clause present in the version that was reviewed?
- Was this policy in effect when the decision was made?
- Was this research protocol finalized before the trial milestone?
- Was this file included in the submitted evidence package?
- Was this article, passage, or dataset present in an AI training corpus?
- Can you prove that a particular item was not included?
- Can you prove integrity without exposing the full record set?

The hard part is not storing the materials. Most organizations already have document management systems, cloud drives, backup tools, audit logs, data warehouses, and internal records.

The hard part is proving a narrow fact about sensitive materials when the people asking the question are not willing to simply trust the organization's internal systems.

That is the provenance gap.

A regulator wants evidence. A court wants evidence. A counterparty wants evidence. A publisher wants evidence. An auditor wants evidence. The producing organization wants to answer precisely without exposing more than necessary.

2. Why Existing Systems Are Not Enough

Existing systems are valuable, but they were not designed to solve adversarial content provenance.

Document management systems can show version history, edits, permissions, timestamps, and user actions. They are useful for internal governance. But they remain controlled by the organization that owns the documents. In a dispute, that creates a trust problem: the system may be accurate, but the counterparty still has to trust the operator.

Cloud storage platforms can show file timestamps and revision history. But cloud metadata is not usually designed to answer narrowly scoped verification questions years later. It can also require exposing files or account-level records that go beyond the specific issue in dispute.

Logs can be changed, rotated, migrated, corrupted, or interpreted differently. Even sophisticated logging and write-once storage can still terminate inside infrastructure controlled by the producer.

The most direct way to prove that something existed is to show it. But that can waive privilege, expose confidential data, reveal trade secrets, violate contractual restrictions, disclose protected health information, compromise source relationships, or reveal proprietary AI training material.

For many organizations, the cost of disclosure is the reason verification is hard in the first place.

3. Competitive Context: What Proveria Is and Is Not

Proveria is not the first system to use hashes, timestamps, signatures, Merkle trees, or public commitments. Those primitives are mature and widely used.

Organizations can already use tools such as OpenTimestamps, RFC 3161 timestamp authorities, content-addressable storage, C2PA-style content credentials, blockchain anchoring, append-only logs, or Merkle structures layered onto an existing document management system.

Those approaches are valuable, but they solve different slices of the problem.

- **Timestamping tools** can prove that a hash existed at a time, but they do not provide a full producer/verifier workflow, receipt language, access control, or passage-level verification model.
- **RFC 3161 timestamp authorities** can support trusted timestamps, but they generally operate at the artifact/hash level and do not define the broader provenance workflow around scoped verification.
- **C2PA and content credentials** are important for media authenticity and content supply chains, but Proveria is focused on private verification of document and dataset membership without requiring disclosure of the underlying set.
- **Content-addressable storage** is useful for integrity and deduplication, but it is not a neutral verification workflow or business evidence package.
- **Sigstore/Rekor and Certificate Transparency-style systems** demonstrate the value of transparency logs over signed claims. Proveria belongs to the same broad design family, but applies the pattern to private content attestations, scoped verification, and business-readable receipts rather than public software artifacts or public certificates.
- **A custom Merkle log inside a DMS** can help internally, but it still usually leaves the trust anchor and workflow inside the producer's own environment.

Proveria's wedge is the combination:

1. **local-first attestation with no plaintext ingestion;**
2. **whole-file and passage-level verification;**
3. **scoped verifier authorization;**
4. **match and no-match result packages;**
5. **human-readable receipts as business artifacts;**
6. **tiered assurance, from self-verifiable public pages to Proveria-signed and optional enterprise-anchored workflows.**

In short: Proveria is not merely a way to timestamp a hash. It is a productized provenance workflow for situations where disclosure is risky, trust is contested, and the result must be usable by legal, compliance, audit, and business stakeholders.

4. The Proveria Approach

Proveria changes the verification model.

Instead of asking the producer to disclose the underlying materials, Proveria allows the producer to create a cryptographic commitment to those materials locally. The commitment represents the set of files and, where supported, passage-level content. The materials stay with the producer. Proveria receives only the cryptographic output needed to support future verification.

The basic workflow is:

1. A producer opens the Proveria desktop app.
2. The producer selects local files or folders.
3. The desktop app hashes and processes the materials locally.
4. The app creates a manifest and Merkle commitment representing the selected set.
5. The manifest is signed by the producer's paired desktop device.
6. Proveria validates the submitted cryptographic metadata.
7. A confirmed attestation becomes immutable.
8. An authorized verifier later checks a specific file hash or passage hash against that attestation.
9. Proveria returns a match or no-match result package and, where applicable, a human-readable receipt.

Proveria is not trying to become another document repository. It is not trying to ingest enterprise records. It is not asking customers to upload sensitive source material into a new system.

It is a provenance layer.

5. A Short Primer: Whole-File Hashing, Shingling, and Future Media Fingerprints

Proveria uses different fingerprinting strategies for different kinds of verification questions.

Whole-file hashing

Whole-file hashing answers a narrow but important question:

Is this exact file the same as a file committed in the attestation?

This is useful for contracts, archives, release packages, evidence files, exported reports, datasets, binaries, images, videos, and other artifacts where exact file identity matters.

Whole-file hashing is precise, but it is brittle by design. If the file is reformatted, converted, OCR'd, tokenized, deduplicated, chunked, or exported differently, the file hash can change.

Shingled passage hashing

Shingling is the bridge between exact file identity and text-content presence.

For text and PDFs, Proveria can normalize text locally, split it into overlapping passage windows, and hash those passage fingerprints. Mechanically, a shingle is an overlapping window of tokens — for example, a protocol-defined sequence of `N` normalized words — hashed individually and committed as part of the attestation. Window size, stride, and normalization rules affect sensitivity, false-positive behavior, and false-negative behavior, so they must be fixed by the protocol version.

The resulting shingle hashes can answer questions like:

Was this clause, paragraph, or passage present in the attested set?

This matters especially for AI training data and legal/document workflows. A training pipeline may tokenize, normalize, deduplicate, or chunk source documents. A publisher's copy of an article may not match the exact bytes seen by a model developer. Whole-file hashing alone will often miss that relationship. Shingled passage fingerprints are designed to make the verification question more content-aware while still remaining cryptographic and scoped.

Shingling is not semantic similarity. It does not prove that a paraphrase, translation, summary, or derivative work was present. It supports exact or normalized passage-level matching under a defined protocol version.

Canonical text normalization is therefore a load-bearing part of the product. Producer and verifier must run the same protocol version and apply the same rules for Unicode normalization, whitespace handling, case folding, punctuation, ligatures, hyphenation, soft hyphens, zero-width characters, line breaks, and paragraph boundaries. If two parties normalize differently, they can get a false no-match even when the human-readable text appears to be the same.

OCR adds another boundary. If the producer attests to a scanned PDF using one OCR engine and the verifier independently OCRs a different copy with different settings, their extracted text may differ. Proveria receipts must therefore distinguish native-text shingle coverage from OCR-derived shingle coverage, and OCR-derived no-match results should be interpreted with that extraction risk in mind.

Future perceptual hashing

For images, audio, and video, exact file hashing is useful but limited. A resized image, transcoded video, or compressed audio file may change bytes while preserving recognizable content.

Future Proveria work may add perceptual image, video, and audio fingerprinting. Those are not part of the core V1 claim. They belong to later protocol work because they introduce different accuracy, false-positive, and false-negative questions.

6. What Proveria Proves — and What It Does Not

Proveria is intentionally precise about its claims.

A Proveria result can prove that a submitted hash matched, or did not match, a specific attestation's committed hash set under a specific protocol version.

In plain terms, Proveria can answer questions like:

- This exact file was part of this attested set.
- This exact file was not found in this attested set.
- This passage-level fingerprint matched content in this attested set.
- This passage-level fingerprint was not found in this attested set.
- This result was checked against this specific attestation.
- This result was produced under this protocol version.
- This result was signed or issued under the applicable trust model.

Inclusion and non-inclusion are not the same primitive

Positive inclusion and negative inclusion should not be treated casually as symmetrical cryptographic claims.

A match can be supported by a Merkle proof path showing that a specific committed leaf belongs to a specific Merkle root. A no-match result requires a different support model. In Proveria V1, no-match results are scoped service-verified statements against the committed leaf set held for a specific attestation. Proveria checks the submitted hash against that attestation's committed leaf set and signs the result package.

That means a no-match result should be read as:

This submitted hash was checked against this specific attestation's committed hash set and was not found.

It should not be read as:

This content does not exist anywhere, was never used, or was absent from all possible corpora.

Future protocol variants may support stronger standalone non-inclusion proofs through sorted Merkle structures, disclosed leaf indexes, or accumulator-style designs. The V1 product claim is narrower and explicit.

What Proveria does not prove

A Proveria result does not, by itself, prove:

- semantic similarity;
- authorship;
- ownership;
- intent;
- whether a paraphrase appeared;
- whether a translation appeared;
- whether a derivative work appeared;
- whether content appeared in some other unattested set;
- whether an organization used the material in a particular way;
- whether the producer attested to every relevant material.

This precision is a feature, not a weakness.

Proveria is strongest when the question is exact and scoped:

Was this file or passage included in this specific committed set, under this specific protocol version?

That scope makes the result defensible.

7. The Receipt as the Business Artifact

The central output of Proveria is not a dashboard. It is a receipt.

A receipt is a durable evidence package that describes the verification result. It can be produced as canonical signed JSON for systems and as a human-readable PDF for business, legal, audit, or regulatory use.

A typical receipt can include:

Receipt

Receipt ID:	vr_2026_05_13_8F29C
Issued at:	2026-05-13T18:42:11Z

What was checked

Producer:	Example Organization
Project:	Q2 Policy Archive
Attestation:	Board-approved policies – May 2026
Attestation status:	Confirmed
Verification type:	Whole-file hash lookup
Submitted hash:	4b9f...91c2

Result	
Result:	Match
Coverage type:	Whole-file
Shingle coverage:	Not applicable
How it was verified	
Protocol version:	proveria-protocol-v1.0
Hash algorithm:	SHA-256
Merkle root:	8ac1...77e4
Who verified	
Verifier identity:	Opposing Counsel LLP / verifier_72B4
Identity assurance:	Email-verified domain + tenant-admin authorization
Verifier access scope:	Authorized for this attestation only
Trust layers	
Signature status:	Proveria-signed
Enterprise anchoring:	Not enabled
Durability	
Verification URL:	https://verify.proveria.example/r/vr_2026_05_13_8F29C
URL behavior:	Confirms receipt signature publicly; attestation details require authorization
QR code:	Included in PDF

For a no-match result, the receipt would identify the scoped attestation checked and state that the submitted hash was not found in that attestation's committed hash set.

The receipt should bind the verification event to a verifier identity or pseudonymous verifier identifier, depending on the access policy. That identity binding makes the receipt more useful in legal, regulatory, and audit settings: the artifact can show not only what was checked, but who was authorized to check it and when the result was issued.

Receipt verification and attestation verification are related but distinct. A verification URL or QR code can allow a third party to confirm that a receipt is authentic and unmodified. Access to private attestation metadata or additional proof details still requires the verifier to be authorized for that attestation.

For shingled results, the receipt should also declare the coverage mode: native text, OCR-derived, mixed, or whole-file only. OCR-derived coverage should include the OCR engine/version where relevant because OCR-derived no-match results carry different evidentiary weight than native-text no-match results.

Receipts matter because high-stakes verification does not end in the product UI. It ends in an email to counsel, a regulator's evidence file, an audit workpaper, a contract dispute, a licensing negotiation, a procurement review, or a board packet.

Proveria turns a technical verification into a business artifact that can travel.

8. Verifier Authorization and Disclosure Control

Proveria's verification model is scoped by design.

For private Team and Enterprise projects, a verifier does not gain access merely because they possess a link. The verifier must be explicitly authorized to verify against a specific attestation.

The V1 model separates roles:

- **Tenant Admins** control disclosure. They authorize consumers/verifiers for specific attestations.
- **Producers** create projects and attestations through the desktop workflow.
- **Consumers or verifiers** can only verify against attestations for which they have explicit access.

This is intentionally stricter than a simple share-link model. A producer can create provenance, but external disclosure is governed by the tenant administrator.

A typical private verification flow is:

1. A Tenant Admin grants a verifier access to a specific attestation.
2. The verifier receives an invitation or access link.
3. The verifier logs in or registers.
4. The verifier sees limited pre-lookup metadata.
5. The verifier submits a hash generated outside Proveria.
6. Proveria returns a match or no-match result package.
7. The lookup and result are audit-logged.

This model lets an organization allow opposing counsel, a regulator, a publisher, a customer, or an auditor to verify one specific attestation without giving that verifier access to the full project, broader workspace, or underlying materials.

Identity assurance

Verifier identity has different evidentiary weight depending on how it is established.

In Team, verifier identity should be treated as platform identity: email verification, account login, tenant-admin authorization, and audit logging. If the verifier uses an organization email domain, the receipt can show that domain-level signal, but it should not imply formal corporate identity verification unless that process actually occurred.

In Enterprise, Proveria can support stronger assurance patterns: SSO or customer IdP authentication, verified organizational domains, manual Proveria-assisted onboarding, or customer-controlled verifier approval workflows. Receipts should state the identity assurance level used, rather than simply displaying a business name as if all names were equally verified.

Authentication at lookup time should be explicit in the audit trail. A receipt is stronger when it can say not only what was checked, but which authenticated verifier account performed the lookup, under what authorization, and at what time.

Authorization lifecycle

Verifier authorization should support expiration and revocation.

If access expires or is revoked, the verifier cannot perform new lookups against that attestation. Previously issued receipts remain valid as historical records of verification events that already occurred. Revoking access does not retroactively invalidate a receipt.

Delegation should be explicit, not implicit. If opposing counsel wants an expert witness to verify the same attestation, the Tenant Admin should grant that expert witness their own verifier access. This preserves identity binding and avoids receipts being passed around as if the original verifier identity applied to someone else.

9. Threat Model and Trust Boundaries

Proveria is designed to reduce trust in internal records and minimize disclosure, but it does not eliminate every risk.

What Proveria is designed to defend against

Post-attestation tampering. Once an attestation is confirmed, the committed metadata and resulting evidence artifacts are treated as immutable. Later changes to source materials do not change the historical attestation.

Disputed existence or timing. A Proveria receipt can help show that a committed representation existed no later than the attestation confirmation time. Producer-side clock manipulation is bounded by Proveria's server-side validation and issuance timestamps, and by optional enterprise anchoring where enabled.

Over-disclosure during verification. Verifiers can check narrow claims without receiving the full corpus or underlying files.

Untrusted internal logs. Proveria creates an evidence layer outside ordinary document management history.

Proveria-only trust in Enterprise workflows. Enterprise can add customer-managed signing and optional anchoring so the trust model does not depend solely on Proveria's platform signature.

Replay of stale submissions. A signed manifest should include protocol version, tenant/project/attestation identifiers, attempt identifier, timestamp, nonce or unique submission identifier, and device signature. Server-side validation rejects duplicate or stale submission attempts so an old signed manifest cannot be replayed as a new attestation.

What Proveria does not defend against by itself

Compromised producer machine before attestation. If the producer's desktop environment is compromised before materials are processed, Proveria cannot know whether the selected files were authentic.

Bad source selection. Proveria proves facts about what the producer selected and attested. It cannot prove the producer selected every relevant document.

False business context. A producer may label a project misleadingly. The cryptographic result concerns the committed set, not the business truth of every label.

Semantic or derivative similarity. Exact file and shingle matching do not prove paraphrase, translation, summarization, or derivative use.

Key compromise without governance. Device keys and enterprise signing keys require revocation, rotation, and operational controls.

Insider threat at Proveria. Team-tier trust depends on Proveria's operational integrity for platform signing, validation, audit controls, and receipt issuance. Enterprise customer-managed signing reduces that dependency for the signing layer by letting the customer's own signing authority stand behind attestations and receipts. Optional anchoring can add external evidence beyond Proveria's platform records.

Cryptographic aging. SHA-256 and Ed25519 are appropriate V1 defaults, but cryptographic systems need agility. Proveria records protocol, hash, signature, canonicalization, and verifier versions so future protocol versions can add or migrate algorithms while preserving historical verification under the rules used at the time of attestation.

Why the boundary matters

A credible provenance product should be clear about what it can and cannot prove. Proveria's trust value comes from scoped, verifiable claims — not from pretending to solve every records, legal, or AI transparency problem at once.

Device-key boundary

The producer desktop app signs submitted manifests with a paired device key. That signature proves the manifest came from an authorized paired desktop profile at submission time. It does not become the sole durable trust anchor for all future verification.

If a producer's laptop is destroyed after an attestation is confirmed, past attestations remain valid because the confirmed attestation, Proveria receipt, and any enterprise customer signatures or anchoring records remain available. The lost device can be revoked and replaced. Producers may have multiple paired devices, each with separate keys, audit history, and revocation status.

Platform survivability

Long-term evidence needs to survive beyond ordinary application availability.

For Team, Proveria-signed receipts should be downloadable as self-contained proof bundles that include the receipt payload, relevant proof data, Proveria signature, and the public verification material needed to validate the signature offline. Proveria should publish and preserve its signing public-key history so a receipt can be checked even if the live API is unavailable.

For Enterprise, customer-managed signing and optional anchoring strengthen survivability further. The customer retains its own signing authority, and anchored records can provide an external timestamped reference outside Proveria's platform records.

The practical distinction is important: Team provides durable Proveria-signed evidence packages; Enterprise can reduce dependence on Proveria's continued operation for long-horizon evidence preservation.

10. Industry Applications

10.1 Legal and Intellectual Property

Legal work is full of provenance questions.

A firm may need to prove that a contract existed in a particular form. A client may need to establish trade secret priority. Opposing counsel may challenge whether a clause was added later. A party may need to prove that a set of documents was produced without alteration. An organization may need to preserve evidence without prematurely disclosing its full contents.

Proveria is valuable because it allows a legal team to preserve proof without turning preservation into disclosure.

Examples include:

- **Contract authenticity.** A firm can attest to executed agreements, drafts, redlines, and closing packages, then later prove that a specific clause or document existed at the relevant time.
- **Trade secret priority.** A company can attest to proprietary designs, methods, technical documents, or business plans without revealing the trade secret itself.
- **Discovery integrity.** A party can attest to a production set and later show that particular files were included, excluded, or unchanged.
- **Privilege-sensitive records.** A legal team can prove that a privileged document existed without exposing the document's contents.
- **Draft history.** A firm can create separate attestations for important negotiation milestones, allowing later proof of what was present at each stage.

Legal evidence posture

Proveria does not determine admissibility. Courts and tribunals decide admissibility under applicable rules.

What Proveria can provide is the kind of structured evidence package that legal teams can use when authenticating digital records: a signed receipt, protocol version, hash, timestamp, verification result, and reproducible process. In U.S. federal practice, that may be relevant to authentication concepts under Federal Rules of Evidence 901 and to self-authenticating electronic evidence mechanisms under Rules 902(13) and 902(14), depending on how the record is prepared, certified, and offered.

The point is not that Proveria makes a legal conclusion. The point is that it gives counsel a more structured technical foundation than a screenshot, email representation, or internal export alone.

10.2 AI Model and Dataset Provenance

AI systems have made provenance questions urgent.

Model developers, dataset builders, content owners, regulators, and enterprise AI buyers increasingly need answers to questions like:

- Was this content included in a training corpus?
- Was this licensed corpus actually used?
- Was this category of protected content excluded?
- Was the evaluation dataset fixed before the benchmark?
- Was a fine-tuning dataset changed after approval?
- Can a model provider substantiate its claims about training data without disclosing the full dataset?

This use case needs careful mechanics.

Whole-file hashing is useful when the thing being verified is a stable artifact: a dataset archive, a release package, a benchmark file, or an exact document. But training pipelines often tokenize, normalize, deduplicate, chunk, filter, and transform content. A publisher's article file may not match the bytes that flowed through a model developer's training process.

That is why shingling matters.

For AI training-data questions, Proveria's stronger V1 path is passage-level verification: normalized text is split into overlapping passage fingerprints and those fingerprints are committed as part of the attestation. A publisher, auditor, or regulator can then check whether specific passages match the attested set, without requiring the model developer to disclose the full corpus.

This still has boundaries. Shingling does not prove paraphrase, translation, summarization, or semantic similarity. It is a content fingerprinting method for exact or normalized passage-level presence under a defined protocol version.

Potential applications include:

- training corpus attestations;
- licensed content audits;
- publisher/model developer disputes;
- EU AI Act readiness workflows, including support for training-data documentation and summary obligations for general-purpose AI models;
- benchmark contamination controls;
- fine-tuning dataset governance;
- model card evidence attachments;
- enterprise AI procurement diligence.

For AI, Proveria's value is not maximum disclosure. It is accountable, scoped disclosure.

V1 should be understood as producer-cooperative. A publisher cannot unilaterally search a model developer's private attestation. The model developer or tenant administrator must authorize the publisher, auditor, regulator, or counterparty to verify against a specific attestation. That makes V1 especially suitable for licensed-content audits, contractual transparency commitments, enterprise procurement diligence, and regulator-facing workflows where the producer agrees to participate. More adversarial publisher-initiated verification requests are better treated as a later workflow, where a verifier can request access and the producer can approve or deny the request.

10.3 Regulated Industries and Audit

Regulated organizations often need to show what records said at a particular point in time.

A bank may need to show the policy in effect when a loan decision was made. A healthcare organization may need to support integrity controls around electronic protected health information. A pharmaceutical company may need to substantiate that a protocol, amendment, or case report form existed at a milestone. A government contractor may need to prove what was submitted and when.

Proveria allows the organization to add a cryptographic evidence layer while keeping the underlying records private.

Examples include:

- **Financial services model and policy governance.** In SR 11-7-style model-risk environments, Proveria can support model documentation provenance by proving what validation reports, policy documents, assumptions, approvals, or review artifacts existed at a particular model review or validation milestone.
- **FDA-regulated electronic records.** For 21 CFR Part 11 environments, Proveria strengthens evidence around record integrity and auditability, including concerns aligned with §11.10(c) record protection and §11.10(e) audit trail expectations.
- **Healthcare integrity controls.** For HIPAA Security Rule workflows, Proveria supports integrity evidence relevant to §164.312(c)(1): protecting electronic protected health information from improper alteration or destruction.

- **Government and control evidence.** Rather than treating FedRAMP as a content-integrity regime, Proveria is better framed against control families such as NIST SP 800-53 AU-10 for non-repudiation and SI-7 for software, firmware, and information integrity.
- **Policy and procedure attestation.** Seal compliance policies when adopted and prove later what the policy said.
- **Clinical trial provenance.** Attest to protocols, amendments, and milestone documents.
- **Manufacturing and quality records.** Prove that production records existed in a particular form at a relevant time.

Proveria is not a compliance certification by itself. It is an evidence layer that can strengthen how organizations demonstrate integrity, timing, and scoped content facts.

10.4 Research and Reproducibility

Research depends on timing, integrity, and priority.

Researchers may need to prove that a hypothesis, protocol, dataset, analysis plan, or negative result existed before publication. They may need to preserve priority without disclosing sensitive or embargoed work. Collaborators may need to resolve contribution disputes. Replicators may need confidence that a dataset has not changed.

Proveria supports research workflows where the record must be fixed before it can be revealed.

Examples include:

- preregistration without immediate disclosure;
- dataset integrity checks;
- protocol and analysis plan attestations;
- priority claims for unpublished work;
- negative result preservation;
- collaboration contribution records;
- reproducibility evidence packages.

The result is a way to separate proving existence from publishing content.

10.5 Journalism and Source Provenance

Journalists and media organizations also face provenance pressure.

A newsroom may need to prove that source materials existed before publication. A subject may claim that a document was altered. A reporter may need to demonstrate that a draft, transcript, image set, or archive existed at a particular time. A small newsroom may need institutional-grade provenance without building its own infrastructure.

Proveria can help newsrooms preserve the integrity of source materials while keeping those materials private.

Examples include:

- source document intake attestations;
- publication-time evidence packages;
- draft milestone records;
- authenticity defense for challenged reporting;
- contributor/source archive provenance.

Proveria does not decide what is true. It helps establish what materials existed and whether specific content was part of an attested set.

10.6 Government Contracting and Procurement

Procurement and government contracting often turn on what was submitted, when it was submitted, and whether the submitted package changed.

Contractors may need to prove the contents of deliverables, compliance materials, proposals, security documentation, technical artifacts, or audit responses. Agencies and oversight bodies may need confidence that records were not altered after submission.

Proveria supports:

- proposal package integrity;
- deliverable attestation;
- oversight response evidence;
- compliance documentation records;
- subcontractor submission provenance;
- dispute support for what was delivered when.

The benefit is especially clear where disclosure must be controlled but integrity still has to be independently credible.

11. Product Model

Proveria V1 is organized around four primary objects:

- **Project** — a workspace or matter containing attestations.
- **Attestation** — a confirmed cryptographic commitment to a selected set of materials.
- **Verification result** — a scoped match or no-match answer against a specific attestation.
- **Receipt** — a durable evidence package representing the result.

Producer attestations are created through the desktop app. The web portal is used for administration, project visibility, consumer authorization, lookup workflows, receipts, and audit visibility.

The product deliberately separates responsibilities:

- Producers create provenance.
- Tenant Admins govern access and disclosure.
- Consumers verify only what they are authorized to verify.
- Proveria validates cryptographic metadata and issues result packages under the applicable trust model.

This separation helps keep the workflow understandable in enterprise settings.

12. Packaging and Trust Levels

Proveria supports multiple trust levels because not every use case requires the same assurance model.

Free

Free is for public, self-verifiable attestations. It supports whole-file hashing and public scoped attestation pages. It does not include private projects, Proveria-signed receipts, shingling, managed verification workflows, or blockchain anchoring.

In Free, self-verification means a verifier can recompute the file hash locally and confirm it against the publicly scoped attestation commitment and producer device-signed manifest data made available for that Free attestation. Free is useful for basic public proof-of-existence and exact-file integrity, but not for private workflows or passage-level AI/content verification.

Trust label:

Self-verifiable public attestation pages — not Proveria-signed

Team

Team is for managed provenance workflows. It supports private projects, shingling, OCR-backed text/PDF processing, Proveria-signed receipts, audit logging, and unlimited fair-use verification.

Trust label:

Proveria-signed managed provenance

Team is available in Starter and Pro levels with the same feature set and different scale limits.

Enterprise

Enterprise is for high-assurance workflows. It includes everything in Team and adds configurable enterprise controls such as complete audit logging, optional customer-managed signing, and optional blockchain anchoring.

Trust label:

Configurable high-assurance provenance with optional customer-managed signing and optional anchoring

The purpose of these tiers is not to create artificial complexity. It is to match assurance level to buyer need.

13. Compliance and Risk Posture

Proveria is designed to support compliance-oriented workflows, but it should be understood precisely.

It is not a substitute for legal advice, regulatory judgment, records governance, or data classification. It is an evidence layer that can strengthen how organizations prove content-level facts.

Relevant compliance themes include:

- **Data minimization.** Proveria avoids collecting plaintext materials.
- **Controlled disclosure.** Verifiers receive scoped results, not entire corpora.
- **Auditability.** Team and Enterprise tiers include audit logging appropriate to their assurance level.
- **Integrity.** Confirmed attestations are immutable within the system.
- **Exportability.** Receipts and proof packages can travel outside the product UI.
- **Retention clarity.** Confirmed artifacts, failed attempts, local drafts, user data, and audit records are treated differently.

For regulated and sensitive environments, Proveria's strongest claim is architectural: it reduces the amount of sensitive material that must move in order to answer a provenance question.

Immutability by tier

Proveria should be explicit about what “immutable” means at each assurance level.

In Team, immutability is enforced by Proveria's platform controls: confirmed attestations are stored as immutable application records, canonical evidence artifacts are written under immutable object keys, updates create new attempts or new attestations rather than mutating confirmed records, and Proveria-signed receipts bind verification results to those records.

In Enterprise, customers that require evidence beyond Proveria's platform commitment can add customer-managed signing and optional external anchoring. Customer-managed signing lets the customer's own signing authority stand behind the attestation or receipt. Anchoring can provide an external timestamped reference that is not solely controlled by Proveria.

This distinction is central to the tier model: Team provides managed provenance under Proveria's trust boundary; Enterprise can reduce reliance on that boundary.

14. Limitations and Honest Boundaries

Proveria is designed to be precise, not magical.

It does not eliminate the need for good records management. It does not determine legal admissibility by itself. It does not prove authorship or ownership. It does not prove semantic similarity. It does not prove how a document was used. It does not guarantee that an organization attested to every relevant record.

What it does is narrower and stronger:

It creates a verifiable record that a defined cryptographic representation of selected materials existed, and later allows scoped verification against that representation.

That precision is what makes it useful.

A vague provenance claim is easy to attack. A scoped, protocol-defined verification result is easier to defend.

15. Roadmap

Proveria V1 focuses on the core provenance loop:

- desktop-first producer workflow;
- local whole-file hashing;
- Team/Enterprise shingling for text/PDF;
- bundled local OCR for scanned PDFs;
- signed manifests;
- server-side validation;
- scoped lookup;
- match/no-match result packages;
- human-readable receipts;
- tiered auditability.

Near-term roadmap items include:

- verification request workflows;
- producer approval flows;
- storage connectors;
- external APIs;
- SDKs;
- webhook integrations;
- richer support tooling;
- tenant-managed templates;
- desktop auto-update;
- billing and paid self-serve upgrades.

Longer-term areas include:

- customer-managed KMS/HSM signing;
- productionized blockchain anchoring;
- advanced retention controls;
- perceptual image, video, and audio hashing;
- semantic similarity workflows;
- enterprise evidence export;
- self-hosted variants;
- multi-region deployment.

Proveria's roadmap is deliberately staged. The first job is to make the core proof workflow reliable, understandable, and defensible.

16. Conclusion

Proveria addresses a problem that is becoming more common across industries: organizations need to prove facts about sensitive digital materials without exposing the materials themselves.

The existing choices are inadequate. Full disclosure is too costly. Internal records are too easy to challenge. Proveria creates a third path: private provenance, scoped verification, and durable receipts.

The value is not simply technical. It is commercial, legal, regulatory, and operational.

A law firm can defend document authenticity without exposing its full file. An AI developer can support training-data claims without publishing its corpus. A regulated company can answer audit questions without over-disclosing records. A research organization can preserve priority without premature publication. A newsroom can defend source integrity without exposing its archive.

Proveria turns provenance from a representation into a checkable result.

It does not answer every question. It answers a specific class of questions extremely well:

What existed, when did it exist, and was this specific content included in the attested set?

For high-stakes digital materials, that is becoming one of the most important questions an organization can answer.